Apple Disease Classification Classifier Using CNN

Accuracy and overfit underfit check:

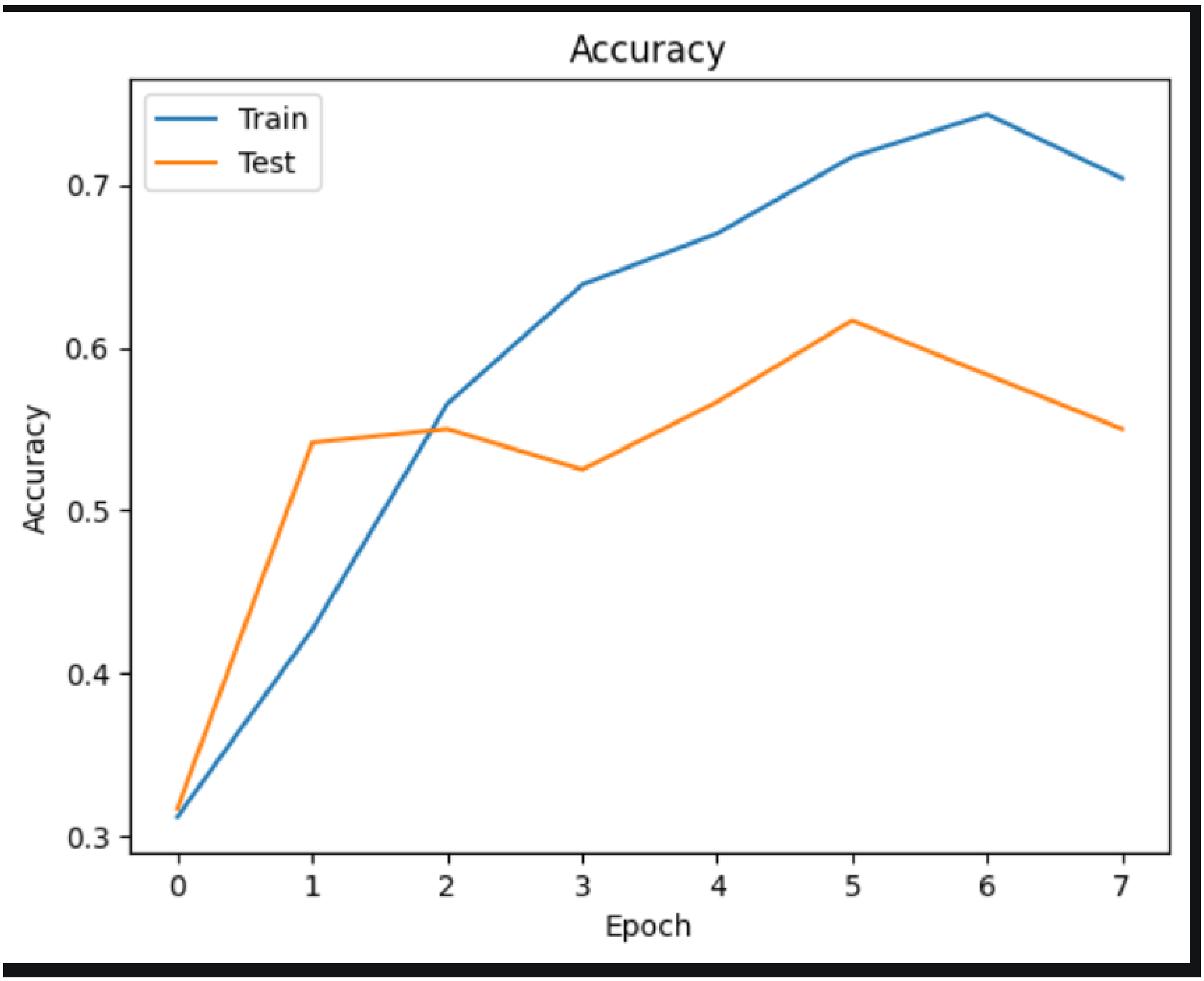
```
12/12 ━━━━━━━━━━━━━━━━━━━━ 11s 814ms/step - accuracy: 0.6204 - loss: 0.9373
4/4 ━━━━━━━━━━━━━━━━━━━━ 3s 704ms/step - accuracy: 0.5500 - loss: 1.0665
Train Accuracy: 62.04%
Test Accuracy: 55.00%
Train Loss: 93.73%
Test Loss: 106.65%
```

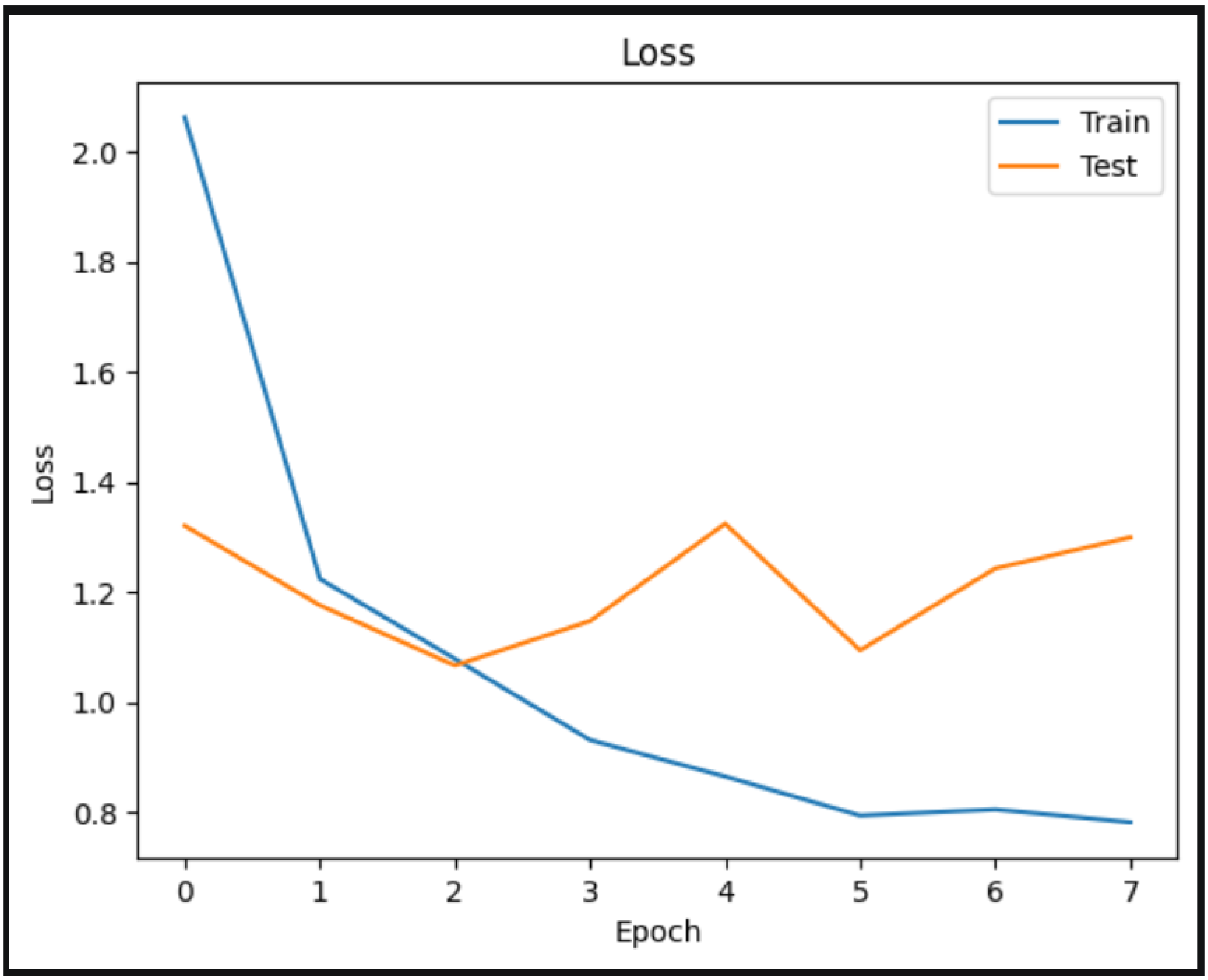
```python
# Step 14: Overfitting Check

if train_acc - test_acc > 0.1:
    print("Overfitting Detected")
elif test_acc > train_acc:
    print("Underfitting Detected")
else:
    print("Good Fit")
```

```
Good Fit
```

Accuracy and Loss Graphs:




Loss
Train
Test
2.0
1.8
1.6
1.4
1.2
1.0
0.8
Loss
0
1
2
3
4
5
6
7
Epoch

Predictions:

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 131ms/step
Predicted Class: Rot_Apple
Confidence: 68.706 %
```

```python
# Step 4: Display Result

import matplotlib.pyplot as plt

plt.imshow(img)
plt.title(f"Prediction: {class_names[predicted_class]} ({confidence*100:.2f}%)")
plt.axis('off')
plt.show()
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 86ms/step
Predicted Class: Scab_Apple
Confidence: 47.174038 %
```

```python
# Step 4: Display Result

import matplotlib.pyplot as plt

plt.imshow(img)
plt.title(f"Prediction: {class_names[predicted_class]} ({confidence*100:.2f}%)")
plt.axis('off')
plt.show()
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 68ms/step
Predicted Class: Normal_Apple
Confidence: 95.996124 %
```

```python
# Step 4: Display Result

import matplotlib.pyplot as plt

plt.imshow(img)
plt.title(f"Prediction: {class_names[predicted_class]} ({confidence*100:.2f}%)")
plt.axis('off')
plt.show()
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 65ms/step
Predicted Class: Blotch_Apple
Confidence: 48.359837 %
```

```python
# Step 4: Display Result

import matplotlib.pyplot as plt

plt.imshow(img)
plt.title(f"Prediction: {class_names[predicted_class]} ({confidence*100:.2f}%)")
plt.axis('off')
plt.show()
```